Supplementary Figure 1

Recombination-filtered phylogenies of candidate surveillance-priority *Klebsiella variicola sequence types*

Source categories

- Human
- Food
- Animal
- Plant
- Environment
- Unknown

Genotype annotation strips

- MDR
- Major virulence
- ESBL
- AMR-virulence
- Carbapenemase
- Absent

Metadata and profile columns

Genome: dataset Genome Name. Country and year: reported metadata; a dash denotes missing information.

Rep: number of distinct PlasmidFinder replicon designations, not a count of complete plasmids.

Key AMR: all reported ESBL/carbapenemase alleles. Virulence: ybt, clb, iuc, iro, rmpADC, rmpA or rmpA2 annotations where present.

Trees and scale convention

Each scale bar denotes recombination-filtered substitutions (Gubbins point-mutation units), not substitutions/site or raw pairwise SNP distances.

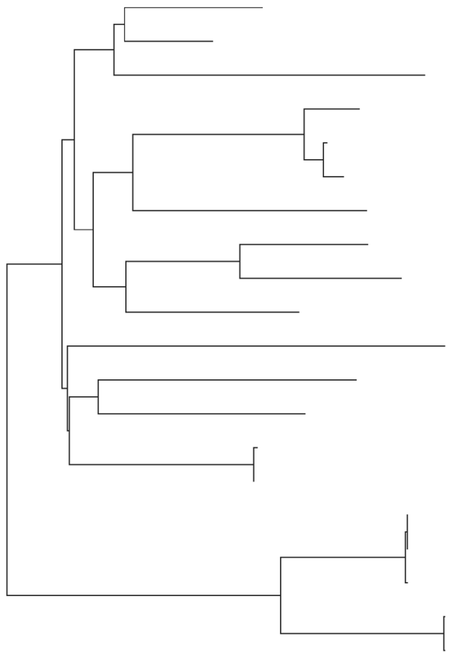
All 299 analysed genomes are shown. Verified identical-core representatives were expanded as zero-length multifurcations.

Branch lengths are linear within each page; scale values differ between STs. No cluster-membership strip is plotted.

Panels A-M follow the main refinement table. The full caption and genotype definitions appear on the final page.

A ST454 | 20 genomes

Clonal resistant expansion

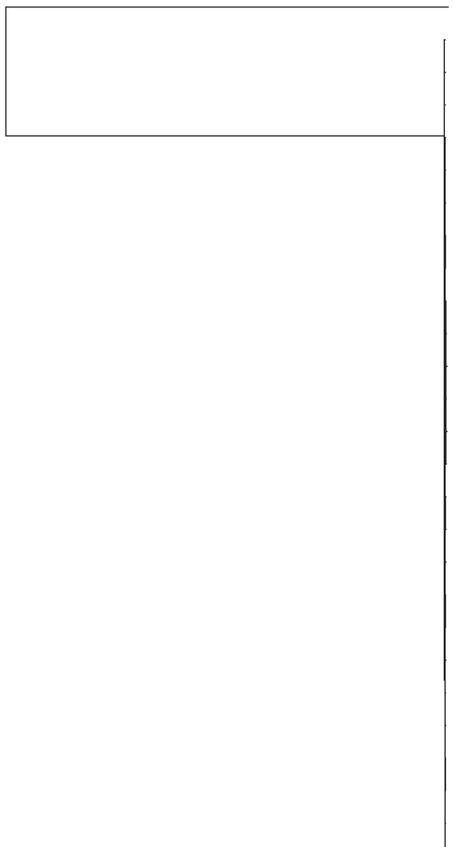


Genome	Source	Country	Year	MDR	ESBL	Carb	Vir	R-V	Rep	Key AMR	Virulence
SAMEA5050053		Italy	2018						0	-	-
SAMEA5050086		Italy	2018						1	-	-
SAMN25041659		Guadeloupe	2019						3	-	-
SAMN35360473		United States	2023						4	-	-
SAMN37671773		Portugal	2019						7	KPC-3	-
SAMN37671771		Portugal	2019						7	KPC-3	-
SAMN26101578		United States	2020						5	-	-
SAMN26764921		Singapore	2020						5	KPC-2, SHV-12	ybt 10; ICEKp4
SAMN50352233		United States	2025						5	KPC-3	-
SAMEA4781917		Italy	2018						6	-	-
SAMN31854198		United States	2022						4	CTX-M-15, NDM-5	iuc 5
SAMEA7198891		United Kingdom	2019						10	-	-
SAMN26419471		United States	2020						6	KPC-3, NDM-1	-
SAMEA113550268		United Kingdom	2018						0	-	-
SAMEA113550269		United Kingdom	2018						1	-	-
SAMN36415720		United States	2023						6	-	-
SAMN36415721		United States	2023						6	-	-
SAMN36415737		United States	2023						7	-	-
SAMN40738337		United States	2024						3	KPC-3	-
SAMN40738340		United States	2024						3	KPC-3	-

200 recombination-filtered substitutions

B ST641 | 27 genomes

Possible cross-source dissemination

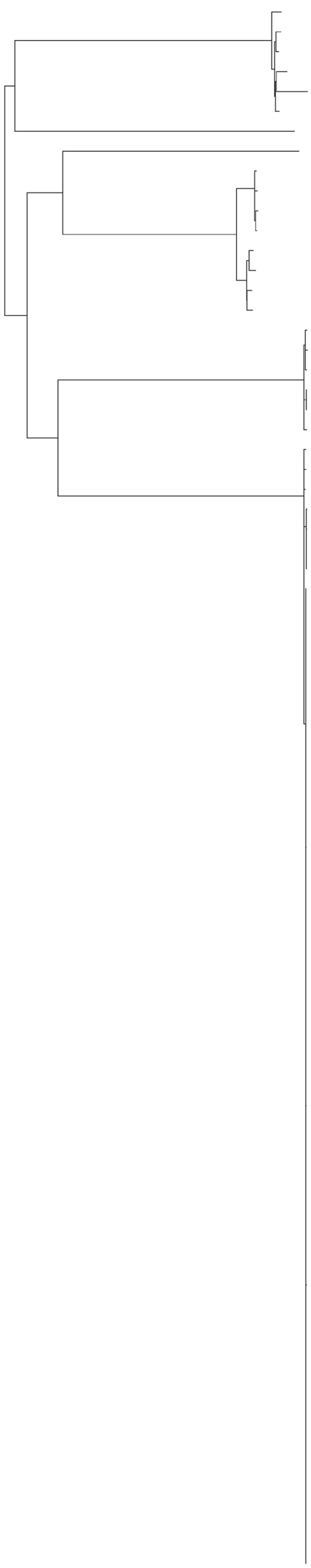


Genome	Source	Country	Year	MDR	ESBL	Carb	Vir	R-V	Rep	Key AMR	Virulence
SAMD00287365		Japan	2018						2	-	ybt 18; ICEKp15
SAMN26976394		United States	2018						6	-	-
SAMEA111505895		Paraguay	2020						4	-	-
SAMN46859435		Australia	2022						1	-	-
SAMN37753163		United States	2023						6	VIM-1	-
SAMN34137692		Canada	2019						4	KPC-3, NDM-1	-
SAMN41955647		United States	2024						2	IMP-18	-
SAMN34030117		-	2023						4	KPC-2	-
SAMN34030118		-	2023						4	KPC-2	-
SAMN26226178		United States	2019						9	KPC-2, SHV-12	-
SAMN26796297		United States	2021						9	KPC-2, SHV-12	-
SAMN26796291		United States	2021						10	KPC-2, SHV-12	-
SAMN26796303		United States	2021						9	KPC-2, SHV-12	-
SAMN26796290		United States	2021						10	KPC-2, SHV-12	-
SAMN26796302		United States	2021						9	KPC-2, SHV-12	-
SAMN26101428		United States	2019						6	-	-
SAMN42513557		United States	2024						6	NDM-7	-
SAMEA113605744		Norway	2018						3	-	-
SAMN38062317		United States	2023						4	NDM-1	-
SAMN36082935		United States	2023						5	NDM-1	-
SAMEA120221852		Germany	2022						2	-	-
SAMEA114784675		Denmark	2019						6	-	-
SAMN32923553		United Kingdom	2021						5	-	-
SAMN28626235		Norway	2019						4	-	-
SAMN28626292		Norway	2019						4	-	-
SAMEA4781725		Italy	2018						3	-	-
SAMN26369074		Australia	2020						5	-	-

20,000 recombination-filtered substitutions

C ST347 | 79 genomes

Clonal resistant expansion

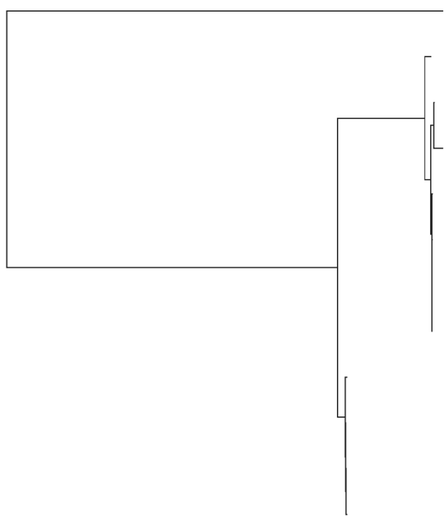


Genome	Source	Country	Year	MDR	ESBL	Carb	Vir	R-V	Rep	Key AMR	Virulence
SAMN48972380	●	Zambia	2022	■	■	□	□	□	4	CTX-M-15	-
SAMEA5049429	●	Italy	2018	□	□	□	□	□	0	-	-
SAMEA4781790	●	Italy	2018	□	□	□	□	□	1	-	-
SAMEA115703377	●	Germany	2023	□	□	■	□	□	7	KPC-2	-
SAMN26743006	●	Germany	2021	□	□	■	□	□	4	KPC-3	-
SAMEA113605433	●	Norway	2018	□	□	□	□	□	0	-	-
SAMD00498095	●	Japan	2020	■	□	□	□	□	1	-	-
SAMN39623894	●	United States	2023	■	■	■	□	□	9	KPC-2, SHV-30	-
SAMEA9454539	●	Italy	2019	□	□	□	□	□	0	-	-
SAMEA5050243	●	Italy	2018	□	□	□	□	□	1	-	-
SAMEA113605206	●	Norway	2018	□	□	□	□	□	3	-	-
SAMEA9455082	●	Croatia	2019	□	□	□	□	□	3	-	-
SAMEA115874794	●	Tanzania, United Republic of	2021	□	□	□	□	□	5	-	-
SAMN26195979	●	Hong Kong	2021	□	□	□	□	□	2	-	-
SAMN40376544	●	United States	2024	□	□	■	□	□	1	KPC-2	-
SAMD00552275	●	Japan	2021	□	□	□	□	□	5	-	-
SAMN48433916	●	Pakistan	2023	□	□	□	□	□	5	-	-
SAMN37365542	●	United States	2023	□	□	■	□	□	2	NDM-5	-
SAMN46910035	●	Slovenia	2024	■	□	■	□	□	5	NDM-1	-
SAMEA113550272	●	United Kingdom	2018	□	□	□	□	□	3	-	-
SAMEA113550273	●	United Kingdom	2018	□	□	□	□	□	4	-	-
SAMEA8218211	●	Germany	2019	□	□	□	□	□	5	-	-
SAMN28626305	●	Norway	2019	□	□	□	■	□	1	-	ybt 18; ICEKp15
SAMN31815268	●	China	2022	■	□	■	□	□	3	NDM-5	-
SAMEA5049722	●	Italy	2018	□	□	□	□	□	0	-	-
SAMN28626269	●	Norway	2019	□	□	□	□	□	0	-	-
SAMN28626237	●	Norway	2019	□	□	□	□	□	0	-	-
SAMN28626254	●	Norway	2019	□	□	□	□	□	0	-	-
SAMN28626290	●	Norway	2019	□	□	□	□	□	0	-	-
SAMN16685854	●	Australia	2018	■	■	□	□	□	9	CTX-M-15	-
SAMN16685858	●	Australia	2018	■	■	□	□	□	9	CTX-M-15	-
SAMN16685853	●	Australia	2018	■	■	□	□	□	9	CTX-M-15	-
SAMN16685847	●	Australia	2018	■	■	□	□	□	8	CTX-M-15	-
SAMN16685848	●	Australia	2018	■	■	□	□	□	8	CTX-M-15	-
SAMN16685852	●	Australia	2018	■	■	□	□	□	9	CTX-M-15	-
SAMN23286217	●	Australia	2018	■	■	□	□	□	8	CTX-M-15	-
SAMN23286220	●	Australia	2018	■	■	□	□	□	8	CTX-M-15	-
SAMN23286228	●	Australia	2019	■	■	□	□	□	8	CTX-M-15	-
SAMN23286232	●	Australia	2019	■	■	□	□	□	8	CTX-M-15	-
SAMN23286231	●	Australia	2019	■	■	□	□	□	8	CTX-M-15	-
SAMN23286233	●	Australia	2019	■	■	□	□	□	6	CTX-M-15	-
SAMN23286223	●	Australia	2019	■	■	□	□	□	8	CTX-M-15	-
SAMN23286226	●	Australia	2019	■	■	□	□	□	7	CTX-M-15	-
SAMN23286229	●	Australia	2019	■	■	□	□	□	8	CTX-M-15	-
SAMN16685856	●	Australia	2018	■	■	□	□	□	8	CTX-M-15	-
SAMN16685857	●	Australia	2018	■	■	□	□	□	9	CTX-M-15	-
SAMN23286206	●	Australia	2020	■	■	□	□	□	5	CTX-M-15	-
SAMN26368947	●	Australia	2020	■	■	□	□	□	5	CTX-M-15	-
SAMN26369157	●	Australia	2020	■	□	□	□	□	5	-	-
SAMN26369154	●	Australia	2020	■	□	□	□	□	5	-	-
SAMN20033514	●	Australia	2018	■	■	□	□	□	8	CTX-M-15	-
SAMN20033516	●	Australia	2018	■	■	□	□	□	8	CTX-M-15	-
SAMN23286222	●	Australia	2019	■	■	□	□	□	8	CTX-M-15	-
SAMN16685850	●	Australia	2018	■	■	□	□	□	8	CTX-M-15	-
SAMN23286225	●	Australia	2019	■	■	□	□	□	8	CTX-M-15	-
SAMN23286227	●	Australia	2019	■	□	□	□	□	8	-	-
SAMN16685851	●	Australia	2018	■	■	□	□	□	8	CTX-M-15	-
SAMN16685846	●	Australia	2018	■	■	□	□	□	8	CTX-M-15	-
SAMN20033534	●	Australia	2019	■	■	□	□	□	8	CTX-M-15	-
SAMN23286218	●	Australia	2018	■	■	□	□	□	7	CTX-M-15	-
SAMN23286230	●	Australia	2019	■	■	□	□	□	6	CTX-M-15	-
SAMN23286208	●	Australia	2020	■	■	□	□	□	6	CTX-M-15	-
SAMN26368982	●	Australia	2020	■	■	□	□	□	6	CTX-M-15	-
SAMN49815546	●	Australia	2020	■	■	□	□	□	6	CTX-M-15	-
SAMN20033526	●	Australia	2018	■	■	□	□	□	6	CTX-M-15	-
SAMN23286216	●	Australia	2018	■	■	□	□	□	8	CTX-M-15	-
SAMN23286221	●	Australia	2019	■	■	□	□	□	8	CTX-M-15	-
SAMN23286213	●	Australia	2018	■	■	□	□	□	8	CTX-M-15	-
SAMN23286214	●	Australia	2018	■	■	□	□	□	8	CTX-M-15	-
SAMN23286224	●	Australia	2019	■	■	□	□	□	6	CTX-M-15	-
SAMN23286219	●	Australia	2018	■	■	□	□	□	6	CTX-M-15	-
SAMN19015836	●	Australia	2018	■	□	■	□	□	7	IMP-4	-
SAMN31262978	●	Australia	2018	■	□	■	□	□	7	IMP-4	-
SAMN32721534	●	Australia	2018	■	■	■	□	□	9	CTX-M-15, IMP-4	-
SAMN49815544	●	Australia	2019	■	■	□	□	□	7	CTX-M-15	-
SAMN16685855	●	Australia	2018	■	■	□	□	□	9	CTX-M-15	-
SAMN23286207	●	Australia	2020	■	■	□	□	□	8	CTX-M-15	-
SAMN26368999	●	Australia	2020	■	■	□	□	□	8	CTX-M-15	-
SAMN23286215	●	Australia	2018	■	■	□	□	□	8	CTX-M-15	-

10,000 recombination-filtered substitutions

D ST981 | 12 genomes

Resistance-virulence convergent sublineage

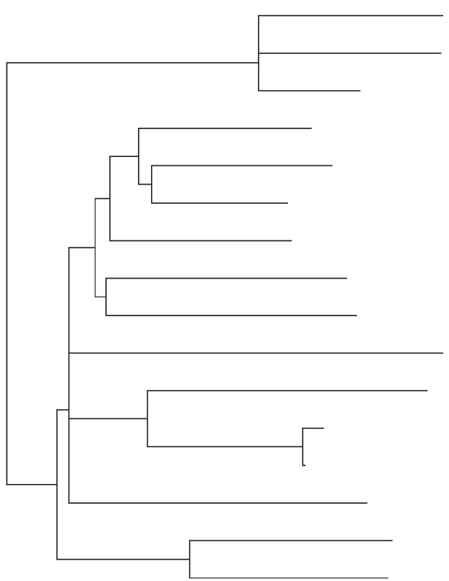


Genome	Source	Country	Year	MDR	ESBL	Carb	Vir	R-V	Rep	Key AMR	Virulence
SAMN29198872	●	Bangladesh	2018	□	□	□	□	□	2	-	-
SAMEA9453267	●	Croatia	2019	□	□	□	■	□	3	-	ybt 18; ICEKp15
SAMN26422391	●	Türkiye	2018	■	□	■	■	■	3	OXA-48	ybt 18; ICEKp15
SAMEA7564872	●	Hungary	2019	■	□	■	■	■	3	OXA-48	ybt 18; ICEKp15
SAMN26422243	●	Türkiye	2018	■	□	■	■	■	3	OXA-48	ybt 18; ICEKp15
SAMN26422267	●	Türkiye	2019	■	□	■	■	■	3	OXA-48	ybt 18; ICEKp15
SAMN26422248	●	Türkiye	2018	■	□	■	■	■	3	OXA-48	ybt 18; ICEKp15
SAMN26422269	●	Türkiye	2018	■	□	■	■	■	3	OXA-48	ybt 18; ICEKp15
SAMEA115100315	●	Switzerland	2022	■	■	□	■	■	4	CTX-M-14	ybt 18; ICEKp15
SAMEA115100308	●	Switzerland	2022	■	■	□	■	■	4	CTX-M-14	ybt 18; ICEKp15
SAMEA115100314	●	Switzerland	2022	■	■	□	■	■	4	CTX-M-14	ybt 18; ICEKp15
SAMEA115100288	●	Switzerland	2022	■	■	□	■	■	4	CTX-M-14	ybt 18; ICEKp15

500 recombination-filtered substitutions

E ST697 | 16 genomes

Recurrent AMR/plasmid acquisition

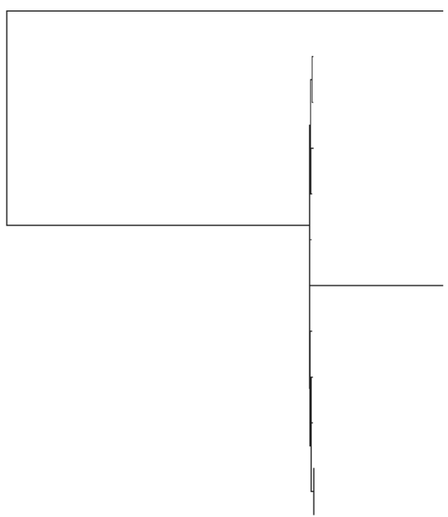


Genome	Source	Country	Year	MDR	ESBL	Carb	Vir	R-V	Rep	Key AMR	Virulence
SAMN33752667	●	United States	2023	■	□	■	□	□	4	KPC-3	-
SAMN41914979	●	United States	2024	□	□	□	□	□	5	-	-
SAMN51424740	●	United States	2025	■	■	■	□	□	5	CTX-M-15, NDM-1	-
SAMEA113550314	●	United Kingdom	2019	□	□	□	□	□	2	-	-
SAMEA7565209	●	Sweden	2019	□	□	□	□	□	0	-	-
SAMN50601332	●	United States	2021	□	□	■	□	□	2	KPC-2	-
SAMN26101511	●	United States	2019	□	□	□	□	□	0	-	-
SAMN32490772	●	United States	2018	□	□	□	□	□	3	-	-
SAMN26369255	●	Australia	2020	□	□	□	□	□	0	-	-
SAMEA113605035	●	Norway	2018	□	□	□	□	□	0	-	-
SAMN38094182	●	United States	2023	■	■	■	□	□	5	NDM-1, SHV-7	-
SAMN26976469	●	United States	2018	■	□	■	□	□	4	KPC-2	-
SAMN26976483	●	United States	2018	■	□	■	□	□	4	KPC-2	-
SAMN25041888	●	Guadeloupe	2019	□	□	□	□	□	0	-	-
SAMN26976335	●	United States	2018	□	□	□	□	□	0	-	-
SAMEA113605539	●	Norway	2018	□	□	□	□	□	2	-	-

200 recombination-filtered substitutions

F ST919 | 12 genomes

Recurrent AMR/plasmid acquisition

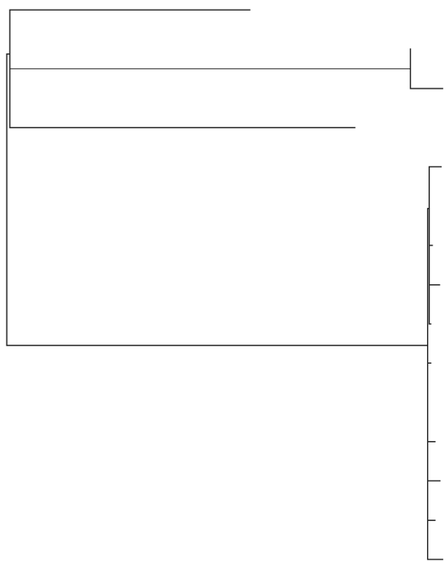


Genome	Source	Country	Year	MDR	ESBL	Carb	Vir	R-V	Rep	Key AMR	Virulence
SAMN52880509	●	Netherlands	2022	■	■	■	□	□	4	CTX-M-15, NDM-1	-
SAMN41914975	●	United States	2024	□	□	□	□	□	4	-	-
SAMN37733889	●	Spain	2022	□	□	■	□	□	8	KPC-2	-
SAMEA113605642	●	Norway	2018	□	□	□	□	□	7	-	-
SAMN34340404	●	Sweden	2021	□	□	■	□	□	9	GES-5	-
SAMEA5684230	●	Italy	2018	□	□	□	□	□	0	-	-
SAMN48972608	●	Tanzania, United Republic of	2021	■	□	□	□	□	3	-	-
SAMN38811463	●	Canada	2020	□	□	□	□	□	0	-	-
SAMEA113605315	●	Norway	2018	□	□	□	□	□	4	-	-
SAMEA8262257	●	Portugal	2019	■	■	□	□	□	5	CTX-M-15	-
SAMN37215957	●	United States	2023	■	□	■	□	□	5	NDM-7	-
SAMN37567195	●	United States	2023	■	□	■	□	□	5	NDM-7	-

20,000 recombination-filtered substitutions

G ST1737 | 15 genomes

Clonal resistant expansion

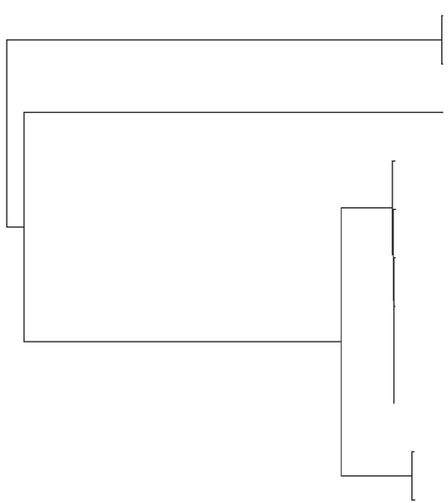


Genome	Source	Country	Year	MDR	ESBL	Carb	Vir	R-V	Rep	Key AMR	Virulence
SAMN34138183	●	Canada	2020	■	□	■	□	□	5	KPC-3	-
SAMN19016368	●	Colombia	2018	■	□	■	□	□	6	KPC-2	-
SAMN19016367	●	Colombia	2018	■	□	■	□	□	6	KPC-2	-
SAMN40296486	●	United States	2023	□	□	□	□	□	4	-	-
SAMN35141348	●	United States	2023	■	□	■	□	□	4	NDM-4	-
SAMN31592763	●	United States	2022	■	■	■	□	□	6	CTX-M-15, NDM-4	-
SAMN31390135	●	United States	2022	■	□	■	□	□	4	NDM-4	-
SAMN31390134	●	United States	2022	■	□	■	□	□	4	NDM-4	-
SAMN31152641	●	United States	2022	■	□	■	□	□	4	NDM-4	-
SAMN31592762	●	United States	2022	■	■	■	□	□	5	NDM-4, SHV-7	-
SAMN31152643	●	United States	2022	■	□	■	□	□	4	NDM-4	-
SAMN31152642	●	United States	2022	■	□	■	□	□	4	NDM-4	-
SAMN31261056	●	United States	2022	■	□	■	□	□	4	NDM-4	-
SAMN31261057	●	United States	2022	■	□	■	□	□	4	NDM-4	-
SAMN31152640	●	United States	2022	■	□	■	□	□	4	NDM-4	-

100 recombination-filtered substitutions

H ST596 | 11 genomes

Recurrent AMR/plasmid acquisition

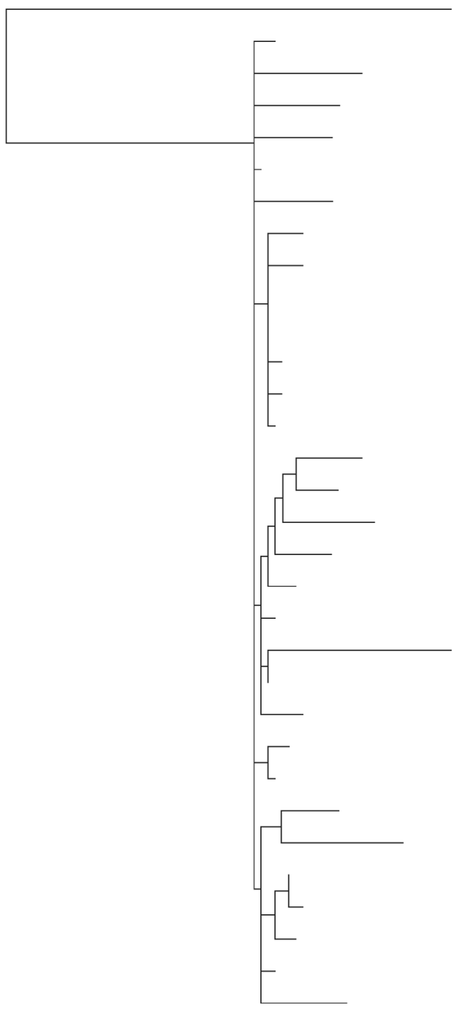


Genome	Source	Country	Year	MDR	ESBL	Carb	Vir	R-V	Rep	Key AMR	Virulence
SAMN35004792	●	Australia	2019	■	□	■	□	□	4	NDM-1	-
SAMEA5050154	●	Italy	2018	□	□	□	□	□	2	-	-
SAMD00502380	●	Japan	2019	■	■	□	□	□	4	CTX-M-3	-
SAMN26976341	●	United States	2018	□	□	□	□	□	0	-	-
SAMN40419759	●	United States	2024	□	■	■	■	■	4	NDM-1, SHV-12	ybt 16; ICEKp12
SAMN48984432	●	United States	2025	□	□	■	□	□	1	KPC-2	-
SAMEA8262857	●	Greece	2019	□	□	□	□	□	0	-	-
SAMEA4781701	●	Italy	2018	□	□	□	□	□	0	-	-
SAMEA4781706	●	Italy	2018	□	□	□	□	□	0	-	-
SAMEA5050052	●	Italy	2018	□	□	□	□	□	1	-	-
SAMEA5050207	●	Italy	2018	□	□	□	□	□	3	-	-

10,000 recombination-filtered substitutions

I ST6219 | 32 genomes

Possible cross-source dissemination

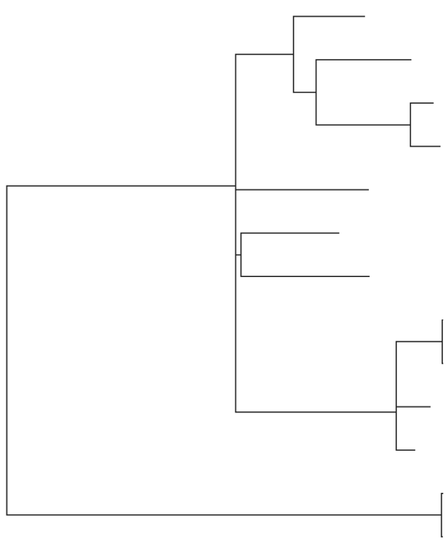


Genome	Source	Country	Year	MDR	ESBL	Carb	Vir	R-V	Rep	Key AMR	Virulence
SAMEA9454398	●	Italy	2019	■	■	□	□	□	2	SHV-12	-
SAMN38450470	●	United States	2023	■	□	■	□	□	2	KPC-3	-
SAMN49914417	●	United States	2025	■	□	■	□	□	2	KPC-3	-
SAMN42886718	●	United States	2024	■	□	■	□	□	2	KPC-3	-
SAMN50265577	●	United States	2025	□	□	■	□	□	2	KPC-3	-
SAMN38120352	●	United States	2023	■	□	■	□	□	2	KPC-3	-
SAMN42337636	●	United States	2024	□	□	■	□	□	2	KPC-3	-
SAMN40918467	●	United States	2024	■	□	■	□	□	2	KPC-3	-
SAMN40918469	●	United States	2024	■	□	■	□	□	2	KPC-3	-
SAMN40918470	●	United States	2024	■	□	■	□	□	1	KPC-3	-
SAMN40918482	●	United States	2024	■	□	■	□	□	2	KPC-3	-
SAMN40918484	●	United States	2024	■	□	■	□	□	2	KPC-3	-
SAMN40918468	●	United States	2024	■	□	■	□	□	2	KPC-3	-
SAMN40918483	●	United States	2024	■	□	■	□	□	2	KPC-3	-
SAMN38120351	●	United States	2023	■	□	■	□	□	2	KPC-3	-
SAMN38335505	●	United States	2023	■	□	■	□	□	2	KPC-3	-
SAMN38120350	●	United States	2023	■	□	■	□	□	2	KPC-3	-
SAMN40161033	●	United States	2024	■	□	■	□	□	2	KPC-3	-
SAMN40375949	●	United States	2024	■	□	■	□	□	2	KPC-3	-
SAMN40217885	●	United States	2024	■	□	■	□	□	2	KPC-3	-
SAMN38120353	●	United States	2023	■	□	■	□	□	2	KPC-3	-
SAMN38450471	●	United States	2023	■	■	■	□	□	3	KPC-3, SHV-30	-
SAMN41612248	●	United States	2024	■	□	■	□	□	2	KPC-3	-
SAMN39832388	●	United States	2023	■	□	■	□	□	2	KPC-3	-
SAMN40174574	●	United States	2024	■	□	■	□	□	2	KPC-3	-
SAMN40643730	●	United States	2024	■	□	■	□	□	2	KPC-3	-
SAMN39255396	●	United States	2023	■	□	■	□	□	2	KPC-3	-
SAMN44254300	●	United States	2024	■	□	■	□	□	4	KPC-3	-
SAMN44083630	●	United States	2024	■	□	■	□	□	4	KPC-3	-
SAMN41472743	●	United States	2024	■	□	■	□	□	4	KPC-3	-
SAMN39997791	●	United States	2024	■	□	■	□	□	3	KPC-3	-
SAMN38710955	●	United States	2023	■	□	■	□	□	2	KPC-3	-

20 recombination-filtered substitutions

J ST197 | 13 genomes

Recurrent AMR/plasmid acquisition

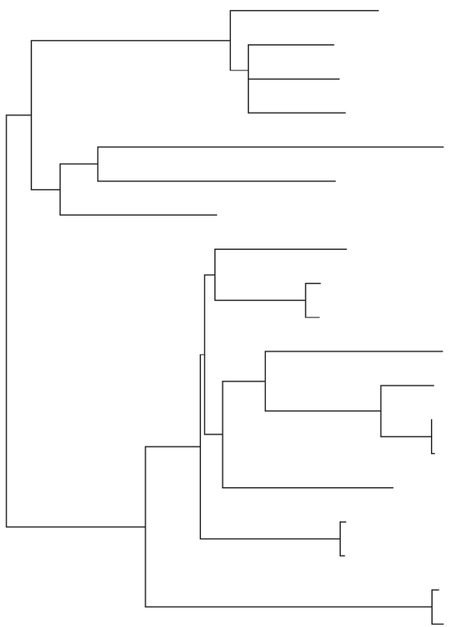


Genome	Source	Country	Year	MDR	ESBL	Carb	Vir	R-V	Rep	Key AMR	Virulence
SAMN28626294		Norway	2019						0	-	-
SAMN26101587		United States	2020						0	-	-
SAMN40296488		United States	2023						4	-	-
SAMN15226415		United States	2019						4	KPC-3	-
SAMN42053668		United States	2024						4	KPC-2, SHV-7	-
SAMN25041724		Guadeloupe	2019						1	-	-
SAMN26369142		Australia	2020						2	-	-
SAMN30630822		United States	2022						6	NDM-7	-
SAMN31713205		United States	2022						6	NDM-7	-
SAMEA8263287		Germany	2019						9	VIM-1	-
SAMD00501569		Japan	2019						4	-	-
SAMN26976427		United States	2018						3	-	-
SAMN26976441		United States	2018						3	-	-

200 recombination-filtered substitutions

K ST616 | 19 genomes

Recurrent AMR/plasmid acquisition

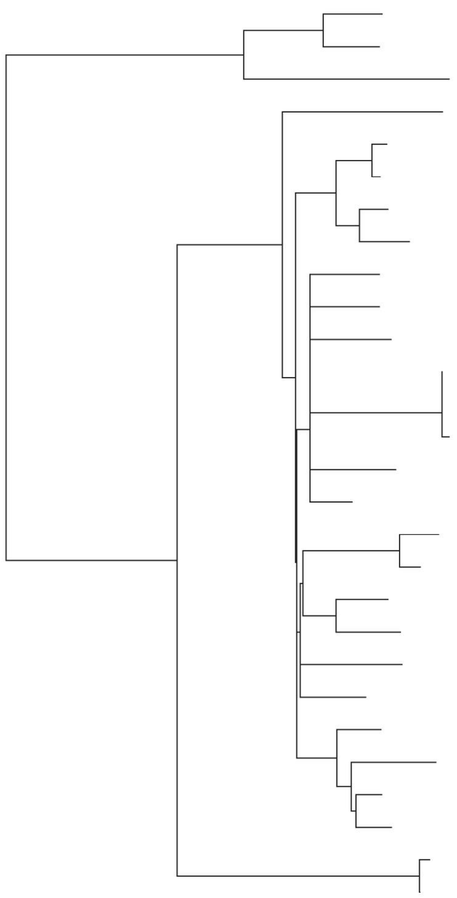


Genome	Source	Country	Year	MDR	ESBL	Carb	Vir	R-V	Rep	Key AMR	Virulence
SAMN44535331	●	United States	2022	■	□	■	□	□	6	KPC-3	-
SAMN42235622	●	United States	2024	■	□	■	□	□	4	KPC-3	-
SAMN38811492	●	Canada	2020	□	□	□	□	□	1	-	-
SAMN40738672	●	United States	2024	□	□	□	□	□	1	-	-
SAMN37798094	●	United States	2023	■	□	□	□	□	8	-	-
SAMN26369200	●	Australia	2020	□	□	□	□	□	6	-	-
SAMEA5049886	●	Italy	2018	□	□	□	□	□	3	-	-
SAMEA9455267	●	Romania	2020	□	□	□	□	□	2	-	-
SAMD00499052	●	Japan	2019	■	■	□	□	□	3	CTX-M-65	-
SAMEA113550267	●	United Kingdom	2018	□	□	□	□	□	1	-	-
SAMD00498818	●	Japan	2020	■	■	□	□	□	3	CTX-M-14	-
SAMD00552220	●	Japan	2019	□	□	□	□	□	2	-	-
SAMN26369067	●	Australia	2020	□	□	□	□	□	3	-	-
SAMN26369195	●	Australia	2020	□	□	□	□	□	2	-	-
SAMN47833567	●	Netherlands	2019	■	■	■	□	□	5	IMP-28, OXA-15	-
SAMEA113550282	●	United Kingdom	2018	□	□	□	□	□	2	-	-
SAMEA113550283	●	United Kingdom	2018	□	□	□	□	□	2	-	-
SAMN26976419	●	United States	2018	□	□	□	□	□	4	-	-
SAMN26976414	●	United States	2018	□	□	□	□	□	4	-	-

100 recombination-filtered substitutions

L ST1562 | 28 genomes

Clonal resistant expansion

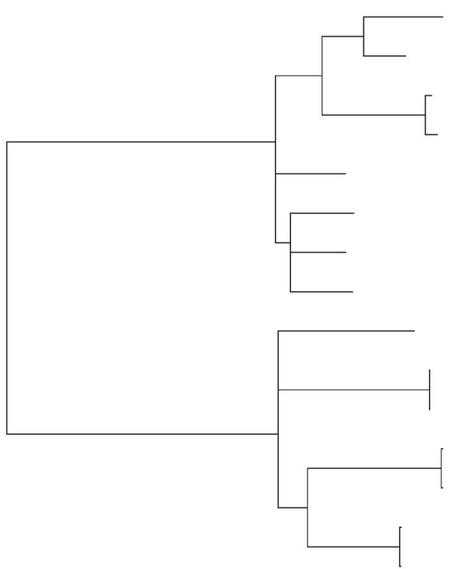


Genome	Source	Country	Year	MDR	ESBL	Carb	Vir	R-V	Rep	Key AMR	Virulence
SAMN26976453	●	United States	2018	□	□	□	■	□	5	-	ybt 0; ICEKp?
SAMEA8263158	●	Bulgaria	2019	□	□	□	■	□	4	-	ybt 0; ICEKp? (truncated)
SAMN47248662	●	Portugal	2021	□	□	■	□	□	6	KPC-3	-
SAMEA113605289	●	Norway	2018	□	□	□	□	□	3	-	-
SAMEA8263700	●	Türkiye	2019	□	□	□	□	□	0	-	-
SAMEA5050115	●	Italy	2018	□	□	□	□	□	0	-	-
SAMEA5050006	●	Italy	2018	□	□	□	□	□	0	-	-
SAMEA11291406	●	Austria	2019	□	□	□	□	□	0	-	-
SAMEA5050180	●	Italy	2018	□	□	□	□	□	1	-	-
SAMN26369094	●	Australia	2020	□	□	□	□	□	5	-	-
SAMN48972204	●	Ethiopia	2021	□	□	□	□	□	2	-	-
SAMN29755192	●	United States	2022	□	□	■	□	□	2	NDM-5	-
SAMN33425791	●	United States	2022	□	□	■	□	□	2	NDM-5	-
SAMN29440613	●	United States	2022	□	□	■	□	□	2	NDM-5	-
SAMEA9455272	●	Romania	2020	■	□	□	□	□	4	-	-
SAMEA9454464	●	Italy	2019	□	□	□	□	□	0	-	-
SAMN49914414	●	United States	2025	□	□	■	□	□	8	KPC-3	-
SAMN57649277	●	United States	2026	□	□	■	□	□	3	KPC-3	-
SAMEA5050055	●	Italy	2018	□	□	□	□	□	3	-	-
SAMN19856757	●	-	2020	□	□	□	□	□	0	-	-
SAMN41914960	●	United States	2024	□	□	□	□	□	3	-	-
SAMD00552194	●	Japan	2019	□	□	□	□	□	2	-	-
SAMN26368929	●	Australia	2020	□	□	□	□	□	2	-	-
SAMEA113605189	●	Norway	2018	□	□	□	□	□	0	-	-
SAMEA114784676	●	Denmark	2019	□	□	□	□	□	2	-	-
SAMEA5049817	●	Italy	2018	□	□	□	□	□	0	-	-
SAMN49947556	●	United States	2025	■	□	■	□	□	2	KPC-2	-
SAMN49096995	●	United States	2025	■	□	■	□	□	2	KPC-2	-

500 recombination-filtered substitutions

M ST208 | 15 genomes

Recurrent AMR/plasmid acquisition



Genome	Source	Country	Year	MDR	ESBL	Carb	Vir	R-V	Rep	Key AMR	Virulence
SAMEA113606015	●	Norway	2018	□	□	□	□	□	0	-	-
SAMEA8263227	●	Bulgaria	2019	□	□	□	□	□	0	-	-
SAMEA111504363	●	Ecuador	2020	■	■	■	□	□	2	CTX-M-12, KPC-2	-
SAMEA111504362	●	Ecuador	2020	■	■	■	□	□	1	CTX-M-12, KPC-2	-
SAMEA5050149	●	Italy	2018	□	□	□	□	□	0	-	-
SAMEA5050156	●	Italy	2018	□	□	□	□	□	2	-	-
SAMEA4781558	●	Italy	2018	□	□	□	□	□	0	-	-
SAMEA114784685	●	Denmark	2019	□	□	□	□	□	0	-	-
SAMEA111504044	●	Croatia	2020	□	□	□	□	□	0	-	-
SAMEA5049523	●	Italy	2018	□	□	□	□	□	1	-	-
SAMEA5049528	●	Italy	2018	□	□	□	□	□	1	-	-
SAMN49075265	●	United States	2025	■	■	■	□	□	2	CTX-M-15, NDM-7	-
SAMN49411769	●	United States	2025	■	■	■	□	□	3	CTX-M-15, NDM-7	-
SAMN26369164	●	Australia	2020	□	□	□	□	□	0	-	-
SAMN26369163	●	Australia	2020	□	□	□	□	□	0	-	-

500 recombination-filtered substitutions

Supplementary Figure 1. Recombination-filtered phylogenies of candidate surveillance-priority *Klebsiella variicola* sequence types.

Panels A-M show ST454, ST641, ST347, ST981, ST697, ST919, ST1737, ST596, ST6219, ST197, ST616, ST1562 and ST208, respectively. The trees include all 299 candidate genomes represented in the recombination-filtered alignments. SAMN26369047 was not included in the ST1562 phylogeny following its earlier exclusion for assembly fragmentation; it remains among the 300 genomes used for ST-level prioritisation. Removed population-QC genomes are not shown. Verified identical core-alignment sequences were restored at zero branch length around their representative, without resolving relationships among indistinguishable tips.

Branch lengths and individual scale bars denote recombination-filtered substitutions in Gubbins point-mutation units, not substitutions per site. These inferred branch lengths are distinct from the observed pairwise SNP counts used for clustering. Scale values differ among panels; each tree is drawn on a linear scale. Root placement follows the retained tree and is not evidence of the direction of spread.

Tip metadata show source category, country, collection year, genotypic MDR, ESBL, carbapenemase, major virulence markers, AMR-virulence convergence and the number of distinct PlasmidFinder replicon designations. MDR denotes an exported resistance-class count of at least three; these reporting categories include separate beta-lactam categories and do not establish phenotypic MDR. ESBL denotes a hit in either acquired ESBL field, including the inhibitor-resistant ESBL field; carbapenemase denotes an acquired carbapenemase hit. Vir denotes a reported yersiniabactin (*ybt*), colibactin (*clb*), aerobactin (*iuc*), salmochelin (*iro*), *rmpADC*, *rmpA* or *rmpA2* annotation. R-V denotes co-carriage of a major virulence marker and MDR, ESBL or carbapenemase status. These annotations do not establish locus functionality or hypervirulence.

Key AMR lists ESBL/carbapenemase alleles; the virulence column gives reported locus annotations. Grey squares indicate feature absence, grey source circles indicate unknown source, and dashes denote missing metadata or no detected gene/locus in the corresponding column. Cluster membership is supplied in the accompanying tables rather than plotted as a separate strip. These within-ST phylogenies refined surveillance prioritisation by assessing resistant expansion, possible cross-source and international clustering, resistance-virulence convergence and dispersed resistance compatible with acquisition or loss. Categories are descriptive and non-exclusive. Close ancestry and matching genomic profiles do not demonstrate direct transmission, independent acquisition events, intact plasmid structures or AMR-gene localisation.